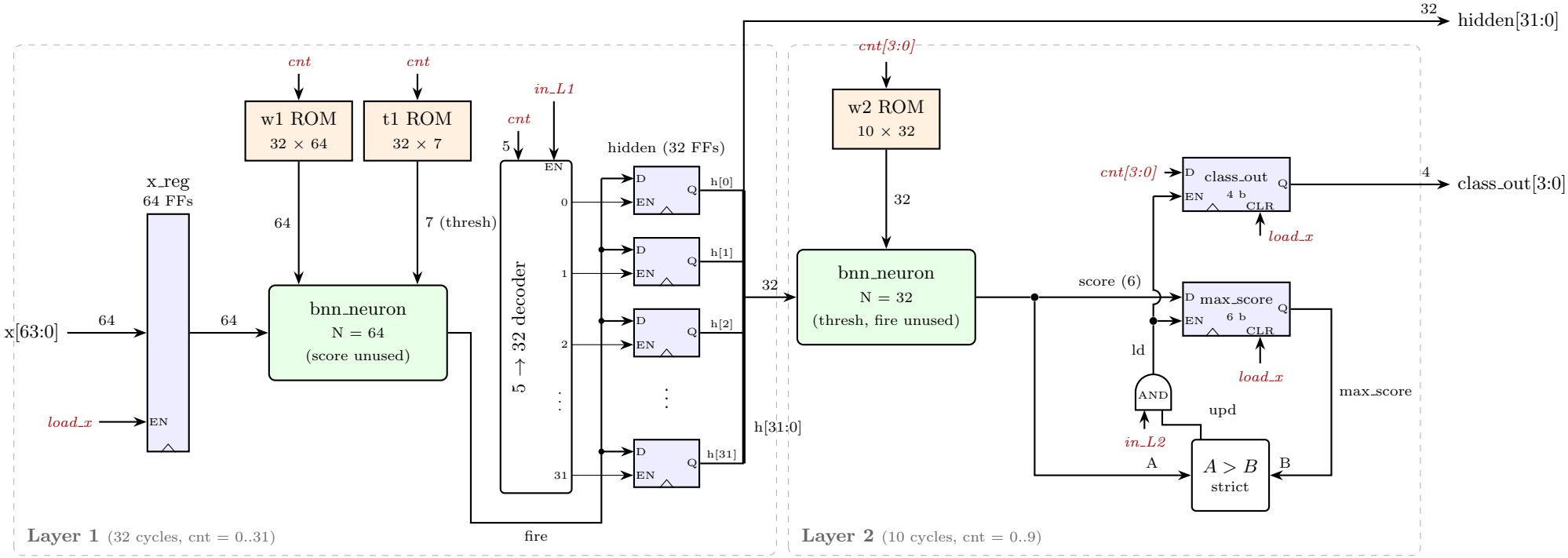
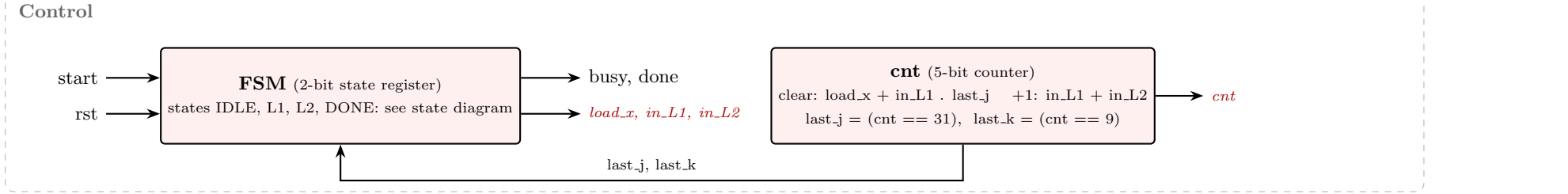
bnn_top, P = 1: datapath and control





Conventions. Flip-flops: D = data in, EN = load enable, Q = out, △ = `posedge clk`, CLR = synchronous clear. Every register also has the synchronous, active-high `rst` (needed for the FSM; optional for data registers). Red italic names are control signals from the FSM / counter, drawn as labels instead of wires. Logic: "." = AND, "+" = OR. `load_x = start . IDLE`; `in_L1`, `in_L2` = state is L1, L2; decoder output `en`$[j]$ = `in_L1`.(`cnt` == j).

Fixes vs. the hand-drawn version. (1) Layer-2 compare is a strict > outside the neuron, so a tie keeps the lower class index (README rule). (2) `max_score` and `class_out` are cleared on every `start`, not only at reset. (3) One FSM and one shared counter replace the two mod counters and the `got` flags; cnt means j in L1 and k in L2. (4) `done` is a one-cycle pulse, raised on the same edge that writes the final `class_out` (L2 → DONE); `hidden` and `class_out` then hold until the next `start`.

Latency. 1 (capture) + 32 (L1) + 10 (L2) + 1 (DONE) ≈ 44 cycles per image.